

Hybrid Machine Learning and CNN Fusion for Facial Emotion Recognition

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Abstract—This project studies seven-class facial emotion recognition using a structured machine learning pipeline. The task is to classify facial images into Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise. Instead of treating deep learning as automatically superior, the project first builds traditional machine learning baselines, then trains a CNN baseline, and finally investigates whether traditional ML methods can complement CNN representations and improve classification performance. The complete pipeline contains four steps: dataset analysis, traditional ML baselines, CNN training with CLAHE comparison, and ML-enhanced CNN fusion. The traditional ML study shows that HOG features are more informative than raw pixels and that HOG with distance-weighted KNN is the strongest traditional baseline. The CNN baseline is selected by validation macro F1 because the dataset is imbalanced. In the final stage, CNN deep features, HOG features, PCA, ML classifiers, probability fusion, stacking, and confusion-aware correction are compared. The best hybrid method improves test macro F1 from 0.5893 for the standalone CNN to 0.6078. The results show that traditional ML methods can provide useful complementary signals for CNN-based facial emotion recognition, especially when evaluation emphasizes balanced performance across all classes.

Index Terms—facial emotion recognition, traditional machine learning, HOG, CNN, probability fusion, stacking, macro F1

I. INTRODUCTION

Facial emotion recognition (FER) is an important computer vision task with applications in human-computer interaction, online education, affective computing, and intelligent tutoring systems. Given a facial image, the goal is to predict one of several emotion categories. In this project, the seven target classes are Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise.

Although facial emotion recognition appears intuitive for humans, it is difficult for machine learning systems. Different expressions can share similar facial muscle patterns. For example, Fear and Surprise may both contain widened eyes and an open mouth, while Sad and Neutral may differ only subtly. In addition, facial emotion datasets are often imbalanced. Majority classes such as Happy may dominate the accuracy score, while minority classes such as Disgust may remain difficult to classify.

The goal of this project is therefore not only to train a CNN model, but also to study how traditional machine learning techniques can be used to understand and improve CNN-based emotion recognition. This design follows the idea that a deep learning model should not be considered superior simply

because it is a deep model. Instead, the model should be compared with traditional ML baselines and analyzed through carefully designed experiments.

The project is organized as a four-step pipeline:

- Step 00 performs dataset analysis.
- Step 01 builds traditional ML baselines.
- Step 02 trains CNN baselines with and without CLAHE.
- Step 03 enhances the CNN using traditional ML representations, probability fusion, stacking, and confusion-aware correction.

II. RELATED WORK

FER has been studied using both traditional machine learning and deep learning methods. Traditional approaches often rely on handcrafted descriptors, while CNN-based methods learn discriminative representations directly from images. However, FER remains difficult because expressions can be ambiguous, images may be low resolution, training data can be limited, class distributions are often imbalanced, and real-world variations such as illumination, pose, and identity can affect facial appearance [2].

The methods used in this project connect to these directions. HOG supports the Step 01 traditional ML baselines by representing local gradient structure [3]. Stacked generalization motivates the Step 03 meta-classifier that combines CNN and ML probability outputs [4]. Grad-CAM supports qualitative explainability by visualizing class-discriminative regions [5]. For future work, larger in-the-wild FER datasets such as AffectNet can test whether the hybrid pipeline generalizes beyond the current benchmark [6].

III. DATASET ANALYSIS

A. Dataset Setting

The dataset is organized into training and test folders, with one subfolder for each emotion class. The project uses seven classes: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise. All steps use a shared configuration file to keep class names, dataset paths, and image settings consistent. The dataset follows the FER2013-style seven-class facial expression recognition setting, in which low-resolution grayscale face images are classified into basic emotion categories [1].

Before model training, Step 00 checks the dataset structure, counts samples in each class, calculates class percentages, and verifies whether unreadable images exist. The dataset contains

TABLE I
CLASS DISTRIBUTION OF THE DATASET

Class	Train	Test	Total	Percentage
Angry	3995	958	4953	13.80%
Disgust	111	436	547	1.52%
Fear	4097	1024	5121	14.27%
Happy	7215	1774	8989	25.05%
Neutral	4965	1233	6198	17.27%
Sad	4830	1247	6077	16.93%
Surprise	3171	831	4002	11.15%
Total	28384	7503	35887	100.00%

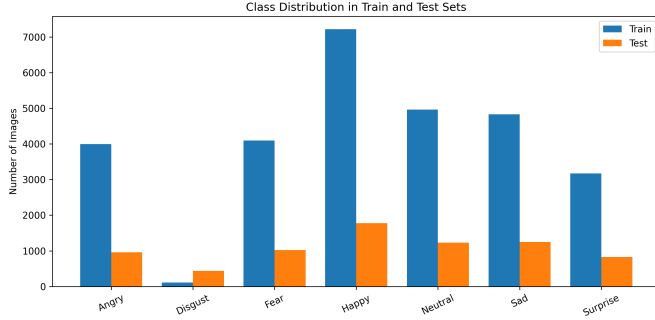


Fig. 1. Class distribution in the training and test sets.

28,384 training images and 7,503 test images, for a total of 35,887 images. No unreadable or corrupted images were found.

Table I shows that the dataset is highly imbalanced. Happy is the largest class, while Disgust is the smallest class. The imbalance ratio between the most frequent and least frequent classes is 16.43:1. This motivates the use of macro F1 as an important evaluation metric, because macro F1 gives equal importance to each class.

A particularly challenging class is Disgust. Although the test set contains 436 Disgust images, the training set contains only 111 Disgust images. This means that the model has very limited training evidence for this class, which may reduce class-wise recall and make macro F1 more informative than accuracy. This also supports the use of class-weighted loss and balanced evaluation metrics.

IV. TRADITIONAL MACHINE LEARNING BASELINES

A. Motivation

Step 01 builds traditional ML baselines before training the CNN. This is useful for two reasons. First, it provides non-deep-learning reference models. Second, it helps identify whether handcrafted features such as HOG contain useful emotion-related information that can later be combined with CNN features.

B. Feature Extraction

Two feature types are compared. The first is a raw pixel baseline, where each 48×48 grayscale image is flattened into

TABLE II
TRADITIONAL ML TEST RESULTS

Feature	Model	Acc.	Macro F1	Weighted F1
HOG	KNN	0.4743	0.4296	0.4583
HOG	Logistic Reg.	0.4100	0.3545	0.3941
HOG	SGD SVM	0.3816	0.3447	0.3728
Raw Pixel	SGD SVM	0.2724	0.2295	0.2627

a vector. The second is the Histogram of Oriented Gradients (HOG), a handcrafted descriptor that summarizes local gradient orientation patterns and has been widely used to represent edge and shape information [3]. HOG is suitable for facial expression recognition because expression changes often appear around eyebrows, eyes, mouth corners, and other local facial structures.

All features are standardized before model training. The original training set is split into training and validation subsets using stratified sampling. Model selection is based on validation macro F1.

C. Traditional ML Models

The following traditional ML baselines are evaluated:

- Raw pixels + SGD linear SVM;
- HOG + Logistic Regression;
- HOG + KNN;
- HOG + SGD linear SVM.

For HOG models, several hyperparameter settings are tested on the validation set. The independent test set is used only for final evaluation.

Table II shows that HOG features are much more useful than raw pixels for traditional ML models. Among the traditional baselines, HOG with distance-weighted KNN achieves the best test macro F1. This result is used as an important design signal in Step 03, where HOG and distance-based ML models are considered as complementary components for CNN enhancement.

V. CNN BASELINE

A. CNN Architecture

Step 02 trains a CNN baseline for seven-class facial emotion recognition. The model uses grayscale 48×48 inputs. The feature extractor contains three convolutional stages. Each stage uses convolutional layers, batch normalization, ReLU activation, max pooling, and dropout. The classifier contains a fully connected layer with 256 hidden units and a final output layer for the seven emotion classes. Table III summarizes the architecture.

B. Preprocessing and Augmentation

All images are resized to 48×48 and normalized to the range $[0, 1]$. Two CNN experiments are compared: one with CLAHE preprocessing and one without CLAHE. CLAHE is used to enhance local contrast in grayscale facial images.

TABLE III
COMPACT CNN ARCHITECTURE

Block	Main Operations
Input	$1 \times 48 \times 48$ grayscale image
Conv Block 1	Conv2D $1 \rightarrow 32$, Conv2D $32 \rightarrow 32$, BN, ReLU, MaxPool, Dropout
Conv Block 2	Conv2D $32 \rightarrow 64$, Conv2D $64 \rightarrow 64$, BN, ReLU, MaxPool, Dropout
Conv Block 3	Conv2D $64 \rightarrow 128$, Conv2D $128 \rightarrow 128$, BN, ReLU, MaxPool, Dropout
Flatten	$128 \times 6 \times 6$
Classifier	Linear($128 \times 6 \times 6$, 256), ReLU, BN, Dropout, Linear(256, 7)

TABLE IV
CNN BASELINE AND CLAHE COMPARISON

Exp.	Val. F1	Acc.	Macro F1	W-F1
CLAHE	0.5786	0.6186	0.5655	0.6102
No CLAHE	0.5818	0.6288	0.5893	0.6231

Training data augmentation includes horizontal flipping, small rotation, brightness and contrast changes, small translation, and mild Gaussian noise. Validation and test data are not augmented.

C. Class Imbalance Handling

The CNN uses class-weighted cross-entropy loss. Class weights are calculated from training label counts, softened using a square-root operation, and clipped within a fixed range. This prevents the minority class weights from becoming too large while still encouraging the model to pay more attention to underrepresented emotions.

D. Training and Model Selection

The CNN is trained using Adam with an initial learning rate of 0.001 and a ReduceLROnPlateau scheduler. The best checkpoint is selected by validation macro F1 rather than validation accuracy. This choice is important because the dataset is imbalanced and the goal is to maintain balanced performance across all emotion classes.

Table IV shows that the CNN without CLAHE obtains the higher validation macro F1 and is selected as the Step 02 best CNN. Its test accuracy is 0.6288 and its test macro F1 is 0.5893. This model is used as the base CNN for Step 03.

VI. ML-ENHANCED CNN

A. Motivation

Step 03 investigates whether traditional ML methods can analyze, complement, or improve the CNN. The CNN is not treated only as an end-to-end black box. Instead, its intermediate representations and probability outputs are used as inputs to several traditional ML methods.

B. CNN Deep Feature Classification

The first experiment extracts the 256-dimensional representation before the final CNN output layer. Logistic Regression, KNN, and SGD linear SVM are then trained on these CNN

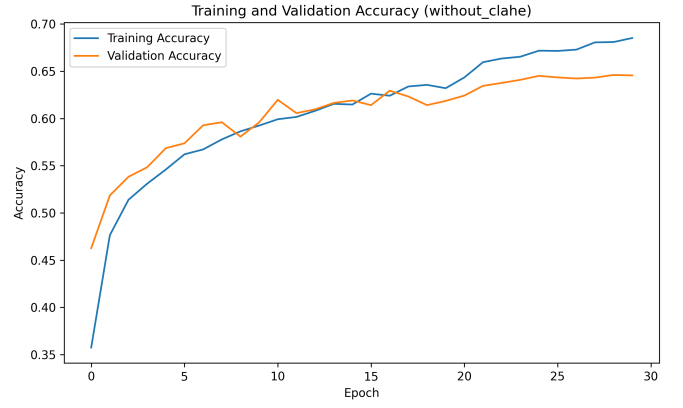


Fig. 2. Training and validation accuracy of the selected CNN without CLAHE.

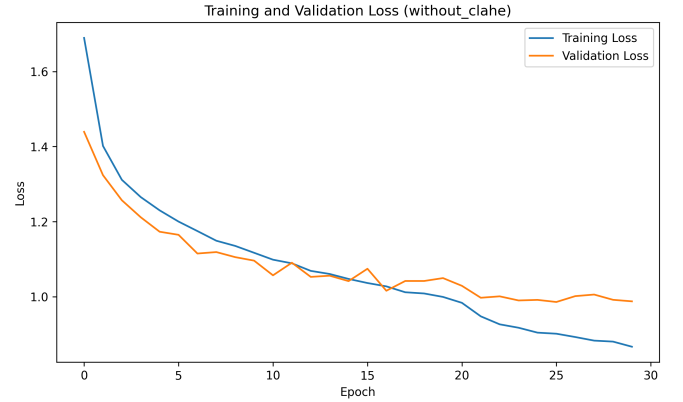


Fig. 3. Training and validation loss of the selected CNN without CLAHE.

deep features. This tests whether the CNN representation is linearly or locally separable for traditional ML classifiers.

C. PCA-Reduced CNN Features

The second experiment applies PCA to the CNN deep features and keeps 64 components. The PCA representation preserves approximately 97.84% of the variance. Traditional ML classifiers are then trained on the reduced features. This experiment checks whether compact CNN representations remain useful for emotion classification.

D. HOG-CNN Feature Fusion

The third experiment combines Step 01 insight with CNN representations. Since Step 01 shows that HOG + KNN is the strongest traditional baseline, HOG features are concatenated with CNN deep features. This creates a feature-level fusion representation that contains both handcrafted gradient information and learned CNN information.

E. Probability-Level Fusion

Feature concatenation alone does not guarantee better performance. Therefore, Step 03 also studies probability-level

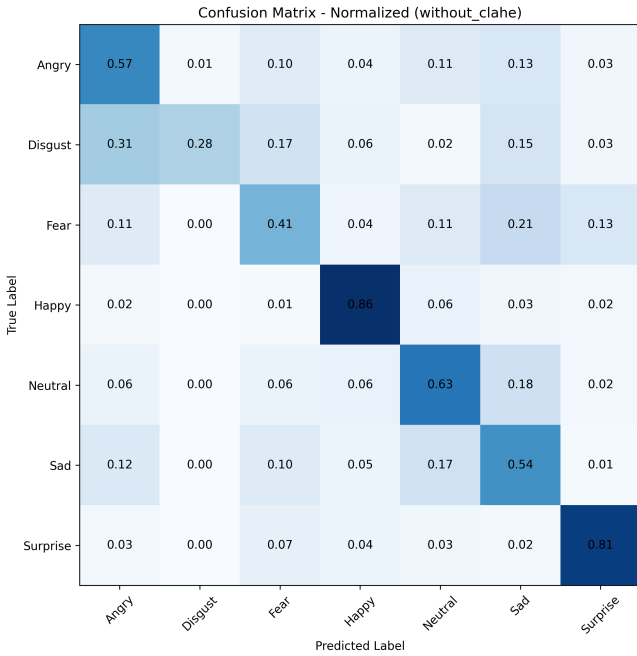


Fig. 4. Normalized confusion matrix of the selected CNN without CLAHE.

fusion. A traditional ML classifier produces class probabilities, and the CNN produces softmax probabilities. The final probability is computed as:

$$P_{\text{final}} = \alpha P_{\text{CNN}} + (1 - \alpha) P_{\text{ML}}, \quad (1)$$

where α is selected on the validation set using macro F1. This allows the final prediction to combine the global CNN decision with the traditional ML classifier's decision.

F. Step 01-Guided Confusion-Aware Correction

The project also includes a confusion-aware correction method. Instead of modifying all CNN predictions, specialist ML classifiers are trained only for frequently confused class groups:

- Angry / Disgust;
- Fear / Sad / Neutral;
- Fear / Surprise;
- Sad / Neutral.

The correction features include CNN deep features, HOG features, CNN probabilities, confidence, top-1/top-2 margin, and entropy. A specialist model is activated only when the CNN is uncertain and its top predictions fall inside a confused group. The final decision uses probability fusion between the CNN and the specialist ML classifier.

G. Stacking

Finally, stacking is used to combine multiple probability outputs, following the idea of stacked generalization where a second-level model learns how to combine the predictions of base models [4]. The stacking features include CNN softmax probabilities and probabilities from ML classifiers trained on

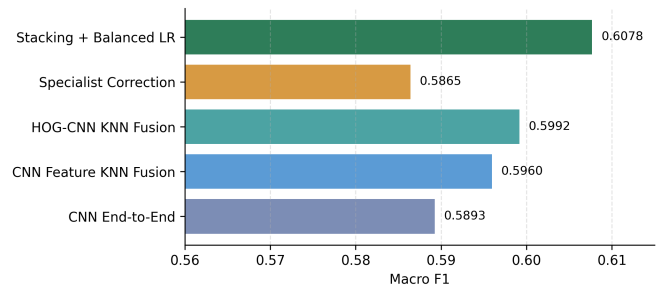


Fig. 5. Macro F1 comparison of the main Step 03 methods on the test set.

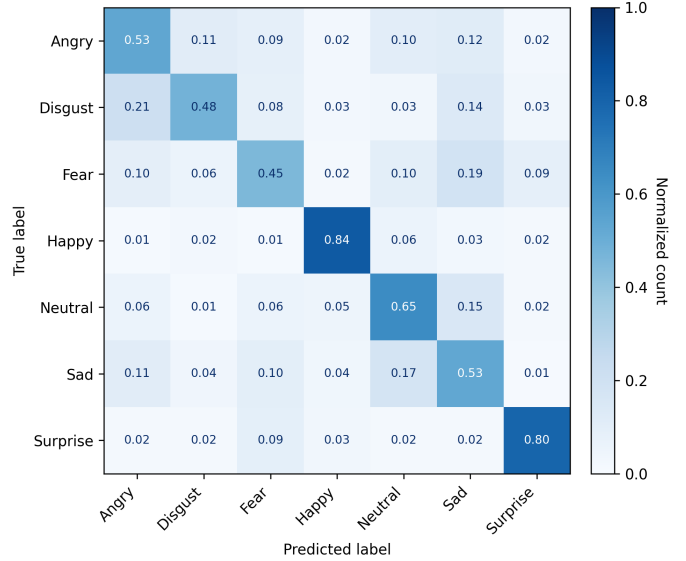


Fig. 6. Normalized confusion matrix of the best Step 03 hybrid model, Stacked CNN-ML Probabilities with Balanced Logistic Regression, on the test set.

CNN deep features, PCA features, and HOG-CNN fusion features. A Logistic Regression meta-classifier is trained on validation probabilities and then evaluated on the independent test set.

Table V summarizes the most important Step 03 results. The best overall test macro F1 is achieved by the stacked CNN-ML probability model with balanced Logistic Regression. It improves test macro F1 from 0.5893 for the standalone CNN to 0.6078, an absolute gain of 0.0185.

VII. RESULT ANALYSIS

A. Traditional ML Findings

The traditional ML baselines show that handcrafted features are meaningful for this task. Raw pixels perform poorly because they do not explicitly describe local facial structure. HOG features perform much better because they capture edge and gradient patterns around important facial regions. HOG + KNN is the strongest traditional baseline, suggesting that local neighborhood structure in HOG space is useful for facial expression classification.

TABLE V
MAIN STEP 03 TEST RESULTS

Category	Method	Accuracy	Macro F1	Weighted F1
Baseline	CNN End-to-End	0.6288	0.5893	0.6231
Probability Fusion	CNN Feature KNN + CNN	0.6388	0.5960	0.6330
Probability Fusion	HOG-CNN KNN + CNN	0.6435	0.5992	0.6357
Correction	Specialist ML Correction	0.6263	0.5865	0.6210
Stacking	Balanced LR Meta-Classifier	0.6388	0.6078	0.6397

B. CNN Findings

The CNN baseline significantly outperforms the traditional ML baselines. This shows that the CNN can learn stronger hierarchical representations from the low-resolution facial images. However, the CNN still has difficulty with minority and visually similar classes. This is why macro F1 is emphasized together with accuracy.

C. Hybrid ML-CNN Findings

The hybrid experiments show that traditional ML methods can improve the CNN when used at the decision level. Simple feature-level fusion is not always enough, because concatenating HOG and CNN features can increase feature dimension without guaranteeing better class boundaries. It may also add noisy or redundant dimensions. Probability-level fusion and stacking are more effective because they combine model decisions rather than only raw feature vectors.

The improvement of stacking suggests that different models provide complementary probability patterns. The CNN softmax captures the end-to-end deep model decision, while ML classifiers trained on CNN features, PCA features, or HOG-CNN fusion features provide alternative decision boundaries. Instead of directly concatenating high-dimensional features, stacking learns how to combine these probability outputs on the validation set. This reduces the risk of adding noisy or redundant feature dimensions and explains why probability-level fusion is more effective than simple feature-level concatenation.

The strongest model is the stacked CNN-ML probability model with balanced Logistic Regression. This model improves macro F1 by 0.0185 over the standalone CNN. Although the absolute improvement is moderate, it is meaningful because macro F1 is sensitive to minority classes and class-level balance.

VIII. ERROR ANALYSIS

Several confusion patterns are observed throughout the experiments. Angry and Disgust can share similar brow and nose-region patterns; Fear and Surprise can both contain widened eyes and open mouths; Sad and Neutral may differ only subtly; and low-resolution images make Fear, Sad, and Neutral harder to separate.

Both Fig. 4 and Fig. 6 are normalized by true class, so they can be compared to examine class-wise recall changes between the CNN baseline and the best hybrid model. The hybrid confusion matrix shows that Happy and Surprise remain

the most reliable classes, while Fear, Angry, Disgust, and Sad are still more difficult. The strongest remaining confusions are conceptually consistent with the CNN baseline: Angry can be confused with Disgust or Sad, Fear overlaps with Surprise and other negative emotions, and Sad can be confused with Neutral. Compared with the standalone CNN baseline, the best hybrid model mainly improves class-wise balance rather than eliminating all visually similar emotion confusions.

The confusion-aware correction experiment was designed based on these patterns. Although it is interpretable, its test macro F1 does not exceed the best stacking model. One reason is that the correction module is activated only for selected uncertain cases, so its coverage is limited. In addition, several confused groups overlap, such as Fear/Sad/Neutral and Fear/Surprise, which makes specialist correction difficult. The Disgust class is also extremely underrepresented in the training set, so the Angry/Disgust specialist has limited examples for learning a stable boundary. Therefore, stacking is more robust because it combines probability information from multiple models globally rather than correcting only selected local cases.

Grad-CAM is used as a supplementary qualitative tool to inspect whether the CNN focuses on expression-related regions. It produces class-discriminative localization maps by using gradients flowing into the final convolutional layer [5]. Fig. 7 shows selected correct and incorrect cases. Correct predictions usually attend to facial regions such as the eyes, eyebrows, and mouth, while incorrect cases may show weaker or less localized attention.

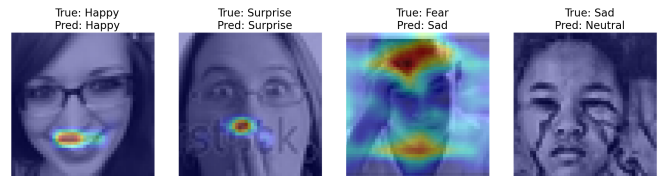


Fig. 7. Grad-CAM examples for selected correct and incorrect CNN predictions.

IX. CONCLUSION

This project implemented a four-step facial emotion recognition pipeline: dataset analysis, traditional ML baselines, CNN baseline training, and ML-enhanced CNN analysis. The traditional ML experiments show that HOG is much more effective than raw pixels and that HOG + KNN is the strongest traditional baseline. The CNN baseline improves performance

substantially, reaching a test macro F1 of 0.5893. The final ML-enhanced CNN experiments show that traditional ML can complement CNN predictions, especially through probability-level fusion and stacking. The best hybrid model improves test macro F1 from 0.5893 for the standalone CNN to 0.6078, corresponding to an absolute gain of 0.0185. However, the improvement is moderate and the model still struggles with visually similar or underrepresented emotions, especially Disgust, Fear, Sad, and Neutral.

Overall, the project demonstrates that traditional ML methods are useful not only as baselines, but also as complementary decision-level components for CNN predictions. CNNs provide strong learned representations, while traditional ML contributes feature analysis, decision-level fusion, and interpretable correction strategies. Future work can further improve the project by tuning the correction mechanism, exploring more compact CNN architectures, applying stronger calibration methods, and testing the approach on larger in-the-wild FER datasets such as AffectNet [6].

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